



Higher order ordinary differential equations

Series solution of differential equations

In many cases, the solutions of linear differential equations with variable coefficients cannot be obtained in closed-form. Basically, a closed-form solution is a general solution to a problem in the form of a closed-form expression, i.e., finite combination of familiar functions known from calculus. The closed-form expressions and solutions are typically restricted to certain operations, i.e., addition, subtraction, multiplication, and division; exponents and logarithms; trigonometric functions and inverse trigonometric functions. Note it that limits and other operations that involve an infinite number of terms or operations are not permitted.

Many of the most important differential equations in applied mathematics, for example, Bessel's equation, Legendre's equation, hypergeometric equation are of this type. Since these type of equations and their solutions play an important role in applied mathematics, so here we consider one of the standard method known as power series method which yields solutions in the form of power series.



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Power series

From calculus, we know that a power series in powers of $(x - x_0)$ is an infinite series of the form

$$\sum_{k=0}^{\infty} a_k (x - x_0)^k = a_0 + a_1 (x - x_0) + a_2 (x - x_0)^2 + \dots$$

where, $a_0, a_1, a_2, \dots$ are constants called coefficients of the series, a_k represents the coefficient of the k^{th} term, x_0 is a constant called the center of the series, and x is a variable. If in particular, $x_0 = 0$, then we obtain a power series in powers of x as

$$\sum_{k=0}^{\infty} a_k x^k = a_0 + a_1 x + a_2 x^2 + \dots$$

Note it that polynomials are also power series, because they have this form.



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Examples of power series are Maclaurin's series:

$$\frac{1}{a-x} = 1 + x + x^2 + x^3 + \dots = \sum_{k=0}^{\infty} x^k \quad (|x| < 1, \text{Geometric series})$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$$

Ordinary and singular points of a differential equation

Consider the following second-order linear homogeneous differential equation with variable coefficients

$$a_0(x)y'' + a_1(x)y' + a_2(x)y = 0, \quad a_0(x) \neq 0. \quad (17)$$



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Dividing the above equation by $a_0(x)$, we can write it in the standard form as

$$y'' + P(x)y' + Q(x)y = 0, \quad (18)$$

where $P(x) = \frac{a_1(x)}{a_0(x)}$ and $Q(x) = \frac{a_2(x)}{a_0(x)}$.

Let $x = x_0$ be a point in some open interval I . A point x_0 is called an ordinary point of equation (18) if $a_0(x_0) \neq 0$ and $P(x)$ and $Q(x)$ are analytic at x_0 . An ordinary point is also known as regular point of the differential equation.

Whereas, a point x_0 is called a singular point of equation (18) if it is not an ordinary point. Thus x_0 is a singular point if $a_0(x_0) = 0$ or if either $P(x)$ or $Q(x)$ fails to be analytic at x_0 .

Example:-

Find the ordinary and singular points of the following differential equation

$$x^3(x-2)^2y'' + 5(x+2)(x-2)y' + 3x^2y = 0.$$



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$$x^3(x-2)^2y'' + 5(x+2)(x-2)y' + 3x^2y = 0.$$

Solution:-

Comparing it with equation (17), we have

$$a_0(x) = x^3(x-2)^2, \quad a_1(x) = 5(x+2)(x-2), \quad a_2(x) = 3x^2.$$

Now $a_0(x) = x^3(x-2)^2 = 0$ implies that $x = 0$ and $x = 2$ which are the singular points. Also, both

$$P(x) = \frac{a_1(x)}{a_0(x)} = \frac{5(x+2)}{x^3(x-2)} \quad \text{and} \quad Q(x) = \frac{a_2(x)}{a_0(x)} = \frac{3}{x(x-2)^2}$$

are not analytic at $x = 0$ and $x = 2$. Whereas, all other real numbers, except 0 and 2, are the ordinary (or non-singular) points of the given differential equation.



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Power series method

The power series method is the standard basic method for solving linear differential equations with variable coefficients. The power series method is used to seek solutions in the form of power series about an ordinary point $x = x_0$. We begin by describing this method for the following differential equation

$$a_0(x)y'' + a_1(x)y' + a_2(x)y = 0, \quad a_0(x) \neq 0. \quad (19)$$

Step-1: Assume the solution of the given differential equation in the form of power series in the powers of $(x - x_0)$ with unknown coefficients, i.e.,

$$y(x) = \sum_{k=0}^{\infty} a_k(x - x_0)^k = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + a_3(x - x_0)^3 + \cdots. \quad (20)$$



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Differentiating equation (20) term-wise, we have

$$y'(x) = \sum_{k=1}^{\infty} k a_k (x - x_0)^{k-1} = a_1 + 2a_2(x - x_0) + 3a_3(x - x_0)^2 + 4a_4(x - x_0)^3 + \cdots, \quad (21)$$

and

$$y''(x) = \sum_{k=2}^{\infty} k(k-1)a_k(x - x_0)^{k-2} = 2a_2 + 6a_3(x - x_0) + 12a_4(x - x_0)^2 + \cdots. \quad (22)$$

Note it that, the series for y' and y'' begin with $k = 1$ and $k = 2$, respectively.



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Step-2: Substitute y , y' and y'' in equation (19) and collect like powers of $(x - x_0)$. Then equate the coefficients of each occurring power of $(x - x_0)$ to zero. This gives relations from which we can determine the unknown coefficients of equation (20).

Step-3: Substitute these coefficients in equation (20) to obtain the series solution of the given differential equation (20).

Example:-

Solve the following differential equation using the power series method about an ordinary point $x = 0$.

$$y'' + y = 0.$$



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Solution:-

Step-1: Let the power series solution of the given differential equation be

$$y(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + \dots \quad (23)$$

Differentiating twice w.r.t. x the above equation, we have

$$y'(x) = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + \dots,$$

$$y''(x) = 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + \dots.$$

Step-2: Substituting $y(x)$ and $y''(x)$ in the given differential equation and then equating the coefficients of various powers of x to zero, we have

$$a_2 = -\frac{a_0}{2} = -\frac{a_0}{2!}, \quad a_3 = -\frac{a_1}{6} = -\frac{a_1}{3!}, \quad a_4 = \frac{a_2}{12} = \frac{a_0}{24} = \frac{a_0}{4!},$$
$$a_5 = -\frac{a_3}{20} = \frac{a_1}{120} = \frac{a_1}{5!}, \dots \quad (a_0 \text{ and } a_1 \text{ are arbitrary}).$$



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Step-3: Substituting the values of $a_2, a_3, a_4, \dots$ in equation (23), we get the following series solution

$$y(x) = a_0 + a_1 x - \frac{a_0}{2!} x^2 - \frac{a_1}{3!} x^3 + \frac{a_0}{4!} x^4 + \frac{a_1}{5!} x^5 + \dots$$

Rearranging the terms, we finally have

$$y(x) = a_0 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right) + a_1 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right).$$

Example:- Solve the following differential equation using the power series method about an ordinary point $x = 0$.

$$(1 - x^2)y'' - 2xy' + 2y = 0.$$

Answer:-

$$y(x) = a_0 \left(1 - x^2 - \frac{1}{3}x^4 - \frac{1}{5}x^6 - \dots \right) + a_1 x.$$

Exercise (Ref. Book No. 1, Problem set 5.1, Question No. 10-14)



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Systems of linear differential equations

Previously, we have discussed the solutions of linear differential equations having one dependent and one independent variable. In many cases, we come across linear differential equations having two or more dependent variables and one independent variable. Such differential equations constitute a system of simultaneous linear differential equations. The complete solutions of such kind of systems can only be obtained if the number of differential equations is equal to the number of dependent variables.

Let x and y are the dependent variables and t is the independent variable. Then the simplest system of differential equations is the system of two first-order linear differential equations in two unknowns x and y and has the form



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$$\begin{aligned}\frac{dx}{dt} &= a_{11}x + a_{12}y + f_1(t), & \text{or} & & (D - a_{11})x - a_{12}y &= f_1(t), \\ \frac{dy}{dt} &= a_{21}x + a_{22}y + f_2(t), & & & -a_{21}x + (D - a_{22})y &= f_2(t),\end{aligned}$$

where all the coefficients a_{11} , a_{12} , a_{21} , and a_{22} are constants. If both $f_1(t)$ and $f_2(t)$ are zero, then the above system of differential equations is called homogeneous otherwise non-homogeneous. To solve the above system of differential equations, sometimes it is convenient to eliminate x or y and then from the resulting equation we can solve for the second variable and then the value of the resulting variable can be obtained.

Example:-

Solve the following system of differential equations



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$$\frac{dx}{dt} = y$$
$$\frac{dy}{dt} = -4x + 4y$$

Solution:- The given system of differential equations can be written as

$$Dx - y = 0 \quad \therefore D = \frac{d}{dt} \quad (24)$$

$$4x + (D - 4)y = 0 \quad (25)$$

Operating $(D - 4)(24) + (25)$, we get

$$D^2x - 4Dx + 4x = 0$$



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$$(D^2 - 4D + 4)x = 0,$$

$$F(D)x = 0,$$

where, $F(D) = D^2 - 4D + 4$. The characteristics (or auxiliary) equation of the above differential equation is

$$F(m) = m^2 - 4m + 4 = 0$$

$$(m - 2)^2 = 0$$

$$m = 2, 2.$$

Since the roots are real and repeated, so we have

$$x = (c_1 + c_2 t)e^{2t}$$

Substitute the value of x in equation (24), we get

$$Dx - y = 0$$

$$y = D(c_1 e^{2t} + c_2 t e^{2t}) = 2c_1 e^{2t} + c_2 e^{2t} + 2c_2 t e^{2t}.$$



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Hence the general solution of the given system of ODEs is

$$x = (c_1 + c_2 t)e^{2t}$$

$$y = 2c_1 e^{2t} + c_2 e^{2t} + 2c_2 t e^{2t}$$

Example:- Solve the following systems of differential equations

$$1) \quad \begin{aligned} \frac{dx}{dt} &= x + y \\ \frac{dy}{dt} &= 4x - 2y \end{aligned}$$

$$2) \quad \begin{aligned} \frac{dx}{dt} - 3y &= 0 \\ \frac{dy}{dt} + z &= 0 \\ \frac{dz}{dt} + y &= 0 \end{aligned}$$

Exercise (Ref. Book No. 1, Problem set 4.6, Question No. 1-15)



Applications of ordinary differential equations

Differential equations are of fundamental importance in science and engineering because, many physical laws and relations appear mathematically in the form of such problems. Here, we will discuss the derivation of differential equations from given physical phenomena. This transition from the given physical phenomenon to a corresponding mathematical problem is called modeling which is of great practical importance, in-particular, to engineers, physicists, and computer scientists.

If we translate a physical phenomenon into a mathematical model then quite often we obtain a set of differential equations along with certain conditions. Whereas, the solution of the system of differential equations provides the solution of the problem.



Applications of ordinary differential equations

Population model, exponential growth or decay (Law of growth)

If relatively small populations (of human, animals, bacteria, etc) are left undisturbed, they often grow according to Malthus's law (named after the English social scientist) which states that the time rate of change of a population (growth or decay) is proportional to the population present at any given time t .

Let $N(t)$ denote the number of a population that is either growing or decaying. The time rate of change of N w.r.t. t is $\frac{dN}{dt}$. According to the law of growth $\frac{dN}{dt}$ is proportional to N , i.e.,

$$\frac{dN}{dt} \propto N.$$

or

$$\frac{dN}{dt} = kN,$$

where k is the constant of proportionality. If the initial population is N_0 ,



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then the initial condition is $N(0) = N_0$. In this way, we have an initial value problem. The solution of this problem is

$$N(t) = N_0 e^{kt}.$$

Note it that, the exponential function e^{kt} increases as t increases for all $k > 0$, and decreases as t decreases. Thus in problems describing growth, such as, population k will have a positive value. Whereas, in problems involving decay, such as, radioactive disintegration, k will yield a negative value. In a similar way to that of population model, we could also model: radioactivity-exponential decay, atmospheric pressure-exponential decay.

Example:-

The population of a town was 1000 a year ago and the present population is 2000. What will be the population of the town after 3 years?



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Solution:-

Let N be the population of the town at any time t . Since the initial population was 1000, so we can say that $N = 1000$ when $t = 0$. If unit of time is year, then at $t = 1$, $N = 2000$. Since the rate of growth $\frac{dN}{dt}$ is proportional to N , so we have

$$\frac{dN}{dt} = kN, \quad (26)$$

where k is the constant of proportionality. The solution of this problem can be obtained by separating the variables as

$$N = Ae^{kt}, \quad A = e^c$$

is the general solution of equation (26).

Using the initial condition that when $t = 0$, $N = 1000$, we get from the above equation

$$A = 1000.$$



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Thus, we have

$$N = 1000e^{kt}.$$

But $N = 2000$ when $t = 1$ implies that $2000 = 1000e^k$, i.e., $e^k = 2$.
Hence, from the above equation, we have

$$N = 1000(2)^t.$$

Since after 3 years total time lapsed will be 4 years and therefore at $t = 4$, the population of the town will be

$$N = 1000(2)^4 \approx 16,000.$$

Example:-

The population of a certain country is known to increase at a rate proportional to the number of people presently living in the country. If after 2 years population has doubled and after 3 years the population is 20,000. Find the number of people initially living in the country?



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Answer:-

The initial population is approx. 7062.

Example:-

If the growth rate of the number of bacteria at any time t is proportional to the number present at t and doubles in 1 week, how many bacteria can be expected after 2 weeks? After 4 weeks?

Answer:-

$$N(t) = N_0 2^t$$

$$N(2) = N_0 2^2 = 4N_0 \text{ (i.e., four times the initial population)}$$

$$N(4) = N_0 2^4 = 16N_0 \text{ (i.e., sixteen times the initial population)}$$

Example:-

A bacterial culture is known to grow at a rate proportional to the amount present. After one hour, 1000 bacteria are observed in the culture and



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after 4 hours 3000. Find an expression for the number of bacteria in the culture at any time t .

Answer:-

The number of bacteria in the culture at any time t is $694e^{0.366t}$.

Example:-

Between 8 a.m. and 9 a.m. the population of a bacterial culture tripled. At what time will the population be 100 times of what it was at 8 a.m.?

Answer:-

Around 12.191 p.m. the population will be 100 times that of the population at 8 a.m.

Example:-

A mold grows at a rate proportional to itself. At the beginning the amount was 2 gms. In 2 days, the amount increased to 3 gms. Find



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Find the amount of mold after 8 days.

Answer:-

After 8 days, the amount of mold is ≈ 10 gms.

Example:-

If ten thousand rupees are invested with annual interest of 10% compounded continuously. What will be the total amount after 5 years?

Answer:-

After 5 years, the capital grows to approx. Rs. 16,500.



Applications of ordinary differential equations

Newton's law of cooling

Newton's law of cooling states that the time rate of change of temperature of the body is proportional to the temperature difference between the body and the surrounding medium.

Let T denote the temperature of the body and T_m denote the temperature of the surrounding medium. Then the time rate of change of the temperature of the body is $\frac{dT}{dt}$, and Newton's law of cooling can be formulated as

$$\begin{aligned} \frac{dT}{dt} &= -k(T - T_m). \\ \text{or} \quad \frac{dT}{dt} + kT &= kT_m, \end{aligned} \tag{27}$$

where we have denoted the constant of proportionality by $-k$ in order to make $\frac{dT}{dt}$ negative, because the temperature of the body decreases with time. Note it that $T > T_m$ in a cooling process. This law was verified experimentally by Newton.



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Equation (27) is a first order linear differential equation in T . The solution of equation (27) is given by

$$T(t) = ce^{-kt} + T_m.$$

If the initial temperature of the body is T_0 , i.e., $T(0) = T_0$. Then from the above equation, we get

$$T_0 = c + T_m, \quad \text{or} \quad c = T_0 - T_m.$$

Hence by substituting the value of c , the temperature of the body at any time t is given by

$$T(t) = (T_0 - T_m)e^{-kt} + T_m.$$

Remarks:-

- 1 In the above formula, it is assumed that T_m remains constant and also the heat flows rapidly enough that the temperature of the body is the same at all points of the body at a given time t .



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- 2 Newton's law of cooling is equally applicable to heating. In this case, it takes the form

$$\frac{dT}{dt} = k(T_m - T).$$

Example:-

A cup of coffee was initially boiling at 120°C . It was placed in air at 20°C . After 10 minutes the temperature of coffee was 80°C . What will be the temperature of coffee after another 20 minutes?

Solution:-

Let T is the temperature of coffee at any time t . Then we know that the differential equation of Newton's law of cooling is

$$\frac{dT}{dt} = -k(T - 20). \quad (28)$$



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If unit of time is minute, then $T = 120$ when $t = 0$ and $T = 80$ when $t = 10$. The solution of equation (28) is given by

$$T - 20 = Ae^{-kt}. \quad (29)$$

Applying the initial condition, i.e., at $t = 0$, $T = 120$, we get $A = 100$. So that we have

$$T - 20 = 100e^{-kt}. \quad (30)$$

Now applying the second condition, i.e., when $t = 10$, $T = 80$. This implies that

$$e^{-k} = \left(\frac{3}{5}\right)^{1/10}.$$

Substituting this value in equation (30), we have

$$T - 20 = 100\left(\frac{3}{5}\right)^{t/10}.$$



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Now the temperature of coffee when $t = 20$ is

$$T - 20 = 100 \left(\frac{3}{5} \right)^{20/10} = 100 \left(\frac{9}{25} \right) = 36,$$
$$T = 36 + 20 = 56.$$

Hence, the temperature of coffee after another 20 minutes is approximately 56°C .

Example:-

A thermometer, reading 5°C , is brought into a room whose temperature is 22°C . One minute later the thermometer reading is 12°C . How long does it take until the reading is practically 22°C , say, 21.9°C ?

Answer:-

Thermometer will take approximately 9.68 minutes until the reading is compatible with room temperature.



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Example:-

Asif Ali, a student of B.Sc engineering, met his neighbour Haji Bashir Hussain after ISHA-Prayer. Early in the morning when he was ready to come to UET, he heard that Haji Bashir was murdered last night. Policemen were gathered there and for investigation purpose, they were interested to find the time of death of Haji Bashir Hussain. Asif Ali is known an expert in the theory of ODEs, resolve their problem. He went in to the room and found that temperature of the body is $80^{\circ}F$ at 8:30 a.m. The temperature of the room is kept constant at $60^{\circ}F$. Two hours later, the temperature of the body dropped to $75^{\circ}F$. Guess the time of death Asif Ali calculated.

Hint:-

Assume that Haji Bashir Hussain was not sick. Then

$$T(0) = T_0 = 98^{\circ}F, \quad T_m = 60^{\circ}F.$$



Applications of ordinary differential equations

Modeling: Electric Circuits

Designing good models is a challenging task the computer cannot do. Therefore, setting up models has become an important task in modern applied mathematics. The best way to gain experience in successful modeling is to carefully examine the modeling process in various fields and applications. Accordingly, modeling electric circuits will be profitable for all students, not just for electrical engineers and computer scientists.

The simplest electric circuit is a series circuit in which there is a source of electric energy supplying an **electromotive force**, such as, a battery or a generator, and a **resistor** which uses energy, for example, an electric light bulb as shown in the following figure.

Figure:- An electric light bulb circuit.



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If we close the switch, a current I will flow through the resistor and this will cause a voltage drop, i.e., the electric potential at two ends of the resistor will be different. This potential difference or potential drop or voltage drop can be measured by a voltmeter.

The basic elements of a series-circuit consist of an inductor, resistor, and capacitor connected in series with a time varying source voltage $E(t)$.

Resistor

A resistor is an electrical component that limits or regulates the flow of electrical current in an electrical circuit. The measure of the extent to which a resistor resists with the flow of current through it is called **resistance**, denoted by R .

Experiments show that the following law holds.



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The voltage drop across a resistor is proportional to the instantaneous current I , and we have

$$E = RI \quad \text{Ohm's law,}$$

where the constant of proportionality R is called the **resistance** of the resistor. The current I is measured in **amperes**, the resistance R in **ohms**, and the voltage E in **volts**. The other two important elements in more complicated circuits are **inductors** and **capacitors**.

Inductor

An inductor is a passive electronic component that stores energy in the form of magnetic field. In its simplest form the conductor consists of a wire loop or coil wound on some suitable material. Basically, an inductor opposes a change in current, having an inertia effect in electricity similar to that of mass in mechanics.



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Whenever current through an inductor changes, i.e., increases or decreases, a counter *e.m.f.* (electromotive force) is induced in it, which tends to oppose this change. Suppose that I denotes the current then the time rate of change of current is given by

$$\frac{dI}{dt}.$$

This produces a counter *e.m.f.* voltage E . Then E is directly proportional to the instantaneous time rate of change of the current I , i.e.,

$$E \propto \frac{dI}{dt} \quad \implies \quad E = L \frac{dI}{dt},$$

where the constant of proportionality L is called the **inductance** of the inductor and is measured in **henry**, denoted by H . time t is measured in **seconds**.



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Capacitor

A capacitor is a passive electronic component of an electronic circuit that has the ability to store energy and opposes any change of voltage in the circuit. Experiments yield the following law.

The voltage drop across a capacitor or condenser is proportional to the instantaneous electric charge Q on the capacitor, i.e.,

$$Q = CE, \quad \text{or} \quad E = \frac{1}{C}Q, \quad (31)$$

where C is called the **capacitance** and is measured in **farads**, the charge Q is measured in **coulombs**. The current I is defined as the instantaneous time rate of change of the charge Q on the capacitor, i.e.,

$$I(t) = \frac{dQ}{dt}.$$



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Thus equation (31) could also be written as

$$E = \frac{1}{C} \int I(t) dt.$$

Kirchhoff's Voltage Law (KVL)

This law states that the sum of all the voltage drops (potential drops) around any closed loop equals the sum of the voltage rises around that loop. In other words, the algebraic sum of voltages around the closed loop is zero.

RL - circuit

An RL-circuit is a circuit that contains a resistance R , the inductance L and an electromotive force (e.m.f.) $E(t)$ as shown in the figure.

Figure:- An RL - circuit.



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We have to obtain the current I at any time t in the circuit assuming R and L to be constants.

We know that the voltage drop across the resistor is

$$E_r = RI.$$

Also, the voltage drop across the inductor is

$$E_L = L \frac{dI}{dt}.$$

By Kirchhoff's voltage law, the sum of two voltage drops must be equal to the electromotive force $E(t)$. Thus, we have

$$L \frac{dI}{dt} + RI = E(t), \quad \text{or}$$
$$\frac{dI}{dt} + \frac{R}{L}I = \frac{E(t)}{L},$$

which is first-order linear differential equation in the dependent variable I .



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RC - circuit

An RC-circuit is a circuit that contains a resistance R , the capacitance C and an electromotive force (e.m.f.) $E(t)$ as shown in the figure.

Figure:- An RC - circuit.

We have to obtain the current I at any time t in the circuit assuming R and C to be constants.

We know that the voltage drop across the resistor is

$$E_r = RI.$$

Also, the voltage drop across the capacitor is

$$E_c = \frac{1}{C} \int I(t) dt.$$



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By Kirchhoff's voltage law, the sum of two voltage drops must be equal to the electromotive force $E(t)$. Thus, we have

$$RI + \frac{1}{C} \int I(t) dt = E(t).$$

To get rid of the integral involved in the above equation, we differentiate it w.r.t. t to get

$$R \frac{dI}{dt} + \frac{1}{C} I = \frac{dE}{dt}, \quad \text{or}$$
$$\frac{dI}{dt} + \frac{1}{RC} I = \frac{1}{R} \frac{dE}{dt},$$

which is first order linear differential equation in the dependent variable I .



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LC - circuit

An LC-circuit is a circuit that contains an inductor of inductance L , a capacitor of the capacitance C and an electromotive force (e.m.f.) $E(t)$ as shown in the figure.

Figure:- An LC - circuit.

We have to obtain the current I at any time t in the circuit assuming L and C to be constants.

We know that the voltage drop across the inductor is

$$E_L = L \frac{dI}{dt}.$$

Also, the voltage drop across the capacitor is

$$E_C = \frac{1}{C} \int I(t) dt.$$



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By Kirchhoff's voltage law, the sum of two voltage drops must be equal to the electromotive force $E(t)$. Thus, we have

$$L \frac{dI}{dt} + \frac{1}{C} \int I(t) dt = E(t).$$

To get rid of the integral involved in the above equation, we differentiate it w.r.t. t to get

$$L \frac{d^2 I}{dt^2} + \frac{1}{C} I = \frac{dE}{dt}, \quad \text{or}$$
$$\frac{d^2 I}{dt^2} + \frac{1}{LC} I = \frac{1}{L} \frac{dE}{dt},$$

which is a second order linear differential equation with constant coefficients.



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RLC - circuit

We now consider the flow of current through an RLC-circuit as shown in the figure. An RLC-circuit consists of a resistor of resistance R , an inductor of inductance L , and a capacitor of capacitance C in series with a time varying voltage source $E(t)$.

Figure:- An RLC - circuit.

The differential equation in RLC-circuit is obtained by considering the following three voltage drops:

The voltage drop across the resistor is

$$E_r = RI.$$

The voltage drop across the inductor is given by

$$E_L = L \frac{dI}{dt}.$$



Applications of ordinary differential equations

The voltage drop across the capacitor is

$$E_c = \frac{1}{C} \int I(t) dt.$$

According to Kirchhoff's voltage law, the sum of three voltage drops must be equal to the electromotive force $E(t)$. Thus, we have

$$RI + L \frac{dI}{dt} + \frac{1}{C} \int I(t) dt = E(t).$$

To get rid of the integral involved in the above equation, we differentiate it w.r.t. t to get

$$R \frac{dI}{dt} + L \frac{d^2 I}{dt^2} + \frac{1}{C} I = \frac{dE}{dt}, \quad \text{or}$$
$$\frac{d^2 I}{dt^2} + \frac{R}{L} \frac{dI}{dt} + \frac{1}{LC} I = \frac{1}{L} \frac{dE}{dt},$$

which is a second order non-homogeneous differential equation with constant coefficients. The general solution of this equation consists of



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a complementary function and a particular integral.

Example:-

An RL-circuit contains a resistor of resistance 8 ohms, an inductor of inductance 0.5 henrys, and a battery of E volts. At time $t = 0$, the current is zero. Find the current at any time $t > 0$ if

$$i) \quad E = 32e^{-8t}, \quad (ii) \quad E = 8te^{-16t}.$$

Solution:-

We know that the differential equation for the current I in an RL-circuit is given by

$$\frac{dI}{dt} + \frac{R}{L}I = \frac{E(t)}{L}. \quad (32)$$



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Since the above differential equation is linear in I , so in-order to solve it, we find the integrator factor as

$$\text{I. F.} = e^{\int \frac{R}{L} dt} = e^{\frac{Rt}{L}}.$$

The general solution of equation (32) is given by

$$I(t) = e^{-\frac{Rt}{L}} \left[\int e^{\frac{Rt}{L}} \frac{E(t)}{L} dt + c \right].$$

Here, $R = 8$ ohms, $L = 0.5$ henries. Therefore, the differential equation (32) becomes

$$\frac{dI}{dt} + 16I = 2E(t), \quad (33)$$

whose solution is given by

$$I(t) = e^{-16t} \left[\int e^{16t} 2E(t) dt + c \right].$$



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i) when $E(t) = 32e^{-8t}$. then we have

$$\begin{aligned} I(t) &= e^{-16t} \left[\int e^{16t} (64e^{-8t}) dt + c \right], \\ &= e^{-16t} \left[\int 64e^{8t} dt + c \right], \\ &= e^{-16t} \left[8e^{8t} + c \right], \\ &= 8e^{-8t} + ce^{-16t}. \end{aligned}$$

Applying the initial condition that when $t = 0$, $I = 0$. From the above equation, we have

$$0 = 8 + c \quad \text{or} \quad c = -8.$$

Thus, the current I at any time t is given by

$$I(t) = 8e^{-8t} - 8e^{-16t}.$$



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Example:-

Find the current $I(t)$ in an RC-circuit at any time t when the circuit has e.m.f. of $10 \sin t$ volts, a resistance of 100 ohms, a capacitance of 0.0005 farad, and no initial charge on the capacitor.

Answer:-

$$I(t) = -\frac{1}{25}e^{-2t} + \frac{1}{50}(2 \cos t - \sin t).$$

Example:-

Find the current $I(t)$ in an LC-circuit when $L = 2$ henrys, $C = 0.005$ farad, $E(t) = 210 \sin 4t$ volts, and $I(0) = 0$, $I'(0) = 0$.

Answer:-

$$I(t) = -5 \cos 10t + 5 \cos 4t.$$



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Example:-

Find the current $I(t)$ in the RLC-circuit when $R = 160$ ohms, $L = 20$ henrys, $C = 0.002$ farad, $E(t) = 37 \sin 10t$ volts, and $I(0) = 0$, $I'(0) = 1$.

Answer:-

$$I(t) = e^{-4t} \left(\frac{3}{26} \cos 3t - \frac{10}{39} \sin 3t \right) + \frac{8}{65} \sin 10t - \frac{3}{26} \cos 10t.$$